

Theorem (Young's Inequality). *Let $a, b \geq 0$ and $p, q > 1$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality holds iff either $a = 0, b = 0$, or $a > 0, b > 0$ and

$$a^p = b^q \quad (\text{equivalently } a^{p-1} = b^{q-1}).$$

Proof. **1. Conjugate exponents.** From $\frac{1}{p} + \frac{1}{q} = 1$ we obtain

$$q = \frac{p}{p-1}, \quad p = \frac{q}{q-1},$$

so that indeed $p, q \in (1, \infty)$.

2. Degenerate cases. If $a = 0$ then $ab = 0$ and the right-hand side equals $\frac{b^q}{q} \geq 0$; thus the inequality holds, with equality only when also $b = 0$. The same argument works when $b = 0$. Henceforth assume $a > 0$ and $b > 0$.

3. Weighted AM–GM. Set

$$\lambda_1 := \frac{1}{p}, \quad \lambda_2 := \frac{1}{q},$$

so that $\lambda_1, \lambda_2 > 0$ and $\lambda_1 + \lambda_2 = 1$. The weighted arithmetic–geometric mean inequality states that for any non-negative numbers x_1, x_2 ,

$$\lambda_1 x_1 + \lambda_2 x_2 \geq x_1^{\lambda_1} x_2^{\lambda_2}, \quad ()$$

with equality iff $x_1 = x_2$. Choosing $x_1 = a^p$ and $x_2 = b^q$ gives

$$\begin{aligned} \frac{1}{p} a^p + \frac{1}{q} b^q &\geq (a^p)^{1/p} (b^q)^{1/q} \\ &= a b. \end{aligned}$$

Rearranging yields Young's inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

4. Equality condition. In (??) equality holds precisely when $x_1 = x_2$, i.e. when

$$a^p = b^q,$$

which is exactly the condition stated in the theorem, together with the degenerate cases treated in step 2.

5. Conclusion. All hypotheses have been used; the desired inequality follows from the weighted AM–GM inequality, and the equality cases follow from the equality condition of that inequality. $\square$ $\square$